

2 (**short**) Air with mass flowrate $\dot{m} = 2.00 \text{ kg s}^{-1}$ initially at $T_1 = 1000 \text{ K}$ and $p_1 = 2.87 \text{ bar}$ flows adiabatically through a mesh in a horizontal pipe. The pipe has constant cross-sectional area $A = 0.02 \text{ m}^2$. The air temperature drops by 28.66 K . Treat air as a perfect gas with $R = 287 \text{ J kg}^{-1}\text{K}^{-1}$, $c_p = 1005 \text{ J kg}^{-1}\text{K}^{-1}$, and $\gamma = 1.4$.

- (a) Calculate the density and speed of the air upstream of the mesh. [2]
- (b) Calculate the speed and density of the air downstream of the mesh and show that the downstream pressure, p_2 is approximately 1.07 bar . [4]
- (c) Sketch this process on a $T - s$ diagram showing relevant lines of constant pressure. [2]
- (d) Calculate the rate of irreversible entropy increase across the mesh. [2]